

EDUCATION

Lovely Professional University
Bachelor of Technology (Computer Science & Engineering)

Jalandhar, India
July 2022 - May 2026

SKILLS SUMMARY

- **Programming:** Python, SQL, Asynchronous Programming
- **Generative AI:** LLMs, Prompt Engineering, Prompt Chaining, Structured Outputs, LLM APIs, Embeddings, NLP
- **Deep Learning:** PyTorch, Transformers, Transformer Architecture
- **Machine Learning:** scikit-learn, Model Evaluation
- **RAG:** Embeddings, Vector Databases, Semantic Search
- **AI Agents:** LangChain, LangGraph, Agentic Workflows
- **Backend:** FastAPI, REST APIs
- **Deployment:** AWS, Docker, Git, CI/CD, Monitoring
- **Software Engineering:** Debugging, Performance Optimization, Testing

WORK EXPERIENCE

- AI Intern | Know AI | [LINK](#)** June 2026 - August 2026
- Built Generative AI and LLM applications using Python, focusing on prompting, evaluation, and performance optimization.
 - Developed Python AI services and REST APIs integrating LLMs, RAG, tool calling, and workflow automation.
 - Designed **RAG pipelines** using embeddings, vector databases, and semantic search for knowledge-grounded retrieval.
 - Developed evaluation pipelines for LLM testing, output validation, error analysis, and reliability measurement.
 - Improved AI system reliability via evaluation, monitoring, debugging, latency optimization, and performance tuning.
- Data Engineer Intern | flickDONE | [LINK](#)** June 2025 - August 2025
- Built scalable Python AI/ML data pipelines processing 50,000+ records for preprocessing and feature engineering.
 - Engineered automated **data preprocessing and validation workflows**, reducing manual verification by 70%.
 - Optimized AWS-based AI/ML pipelines, improving execution speed by 25% through performance tuning.
 - Developed reusable AI workflows for document processing, LLM applications, API integrations, and enterprise automation.
 - Collaborated with teams to build scalable AI/ML workflows and improve system reliability and deployment.

PROJECTS

- Ledgerly — Multi-Tenant SaaS Billing Platform | [LIVE](#)** April 2026 - September 2026
- Built a multi-tenant SaaS platform using Next.js, TypeScript, NestJS, PostgreSQL, Prisma, and JWT.
 - Implemented tenant-isolated customer and invoice workflows with **RBAC and transactional data operations**.
 - Built **Stripe payment workflows and webhook processing** with idempotency and transactional updates.
 - Added **rate limiting, health checks, and secure error handling** for production-grade reliability.
 - Developed **174 automated tests across 21 suites** covering authentication, billing, payments, and tenant isolation.
- LangGraph Workflow Orchestrator — Agentic AI Platform | [LIVE](#)** August 2025
- Built and deployed a stateful **Agentic AI platform** using Python, LangGraph, LangChain, FastAPI, and Gemini.
 - Designed multi-step agentic workflows with state management, task decomposition, and error recovery.
 - Developed modular components for **task decomposition, tool calling, function calling, and agent coordination**.
 - Exposed AI workflows through FastAPI REST APIs, enabling scalable deployment and integration with external services.
 - Improved workflow reliability through **checkpointing, testing, failure handling, and monitoring**.
- Production-Grade RAG Engine | [LIVE](#)** June 2024 - August 2024
- Built an end-to-end production-grade RAG system using Python, FastAPI, LangChain, Qdrant, and AWS.
 - Developed document preprocessing, chunking, embeddings, indexing, retrieval, and response generation pipelines.
 - Implemented **semantic vector search with Qdrant** for low-latency, knowledge-grounded context retrieval.
 - Added reranking and retrieval evaluation to improve retrieval quality, context relevance, and AI response performance.
 - Improved AI service reliability through testing, monitoring, fallback handling, and error analysis.

CERTIFICATES

- Python, Data Science & ML (Cipherschools) | [CERTIFICATE](#)** August 2024
- Built machine learning models using PyTorch and applied ML techniques for model development and evaluation.
 - Developed **Dockerized environments** supporting reproducible experimentation and deployment
- Generative AI with Large Language Models (Coursera) | [CERTIFICATE](#)** March 2024
- Applied prompt engineering techniques for production LLM applications.
 - Built **retrieval workflows, AI output evaluation, and RAG pipeline** performance validation.